

PATRICK McNAUGHTON

Blue Bell, PA

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Education

Purdue University

Bachelor of Science in Computer Science

Aug. 2026 – May 2029

West Lafayette, IN

Experience

FRC Team 341 (Miss Daisy)

2022 – 2026

Captain, Programming Lead, Strategy Lead

Ambler, PA

- Captained a 60-person FRC team; achieved a peak global rank of 20th out of 3,274 with 10 event wins and 24 awards.
- Owned the robot requirements process, coordinating electrical, mechanical, programming, and strategy leads to translate the season's optimal strategy into a concrete 'robot will-do list' of engineering requirements.
- Led robot software development as lead developer across junior and senior seasons, directing contributors and mentoring younger students through presentations, demonstrations, and video tutorials.
- Drove prematch strategy, analyzing scouting data to build game plans with alliance partners, reviewing match film with the drive team, and representing the team in alliance selection.
- Redesigned the team's engineering process by creating a season-long subsystem Gantt schedule, implementing an alpha-beta-competition cycle across three successive robots, and launching an offseason robot program to test new techniques, products, and strategies.

Projects

Kinemo | *Python, MediaPipe, OpenCV, NumPy* | **1st Place Overall, PennApps XXVI** | [GitHub](#) | [Devpost](#)

2025

- Built a webcam-only motion-controlled gaming platform in 48 hours with people I had never met before using real-time body tracking, winning 1st Place Overall at PennApps XXVI.
- Engineered a real-time computer-vision pipeline using OpenCV and MediaPipe pose estimation to translate player body motion into responsive game controls.

2026 Robot Code | *Java, WPILib* | github.com/Team341/FRC2026-Public

2026

- Implemented and tuned shoot-on-the-move using a constant time-of-flight and interpolated flywheel and turret setpoints, enabling accurate scoring while translating and rotating anywhere across the field.
- Fused latency-compensated AprilTag poses from a turret-mounted camera with wheel odometry using a Kalman filter for robust field localization.
- Controlled the turret with PID and angular-rate feedforward, compensating for robot translation and rotation to stay locked on quickly moving targets.
- Extensively automated robot actions, reducing manual controls to translation, rotation, and one score button.

2025 Robot Code | *Java, WPILib* | github.com/Team341/FRC2025-Public

2025

- Developed state-machine control for a 5-degree-of-freedom robot, winning an Innovation in Control Award.
- Coordinated interdependent mechanisms through defined states and transitions, automatically sequencing complex motion while preventing unsafe configurations.
- Implemented auto-alignment to drive automatically to scoring locations based on robot state and driver input, reducing scoring cycle times, and resulting in a 40.7% increase in average points scored per match

Technical Skills

Languages: Java, Python, C

Developer Tools: vscode, neovim, tmux, Git/GitHub, Linux/WSL2, macOS, windows

Volunteering

Calvary United Methodist Church

2020 – Present

Audio/Visual Technician

Blue Bell, PA

- Run audio, video, and livestream production for weekly services and events, managing setup, mixing, and troubleshooting.

Sparky's Robot World / STEM Sprouts

2023 – 2026

STEM Outreach Volunteer

Blue Bell, PA

- Brought demonstration robots to local events to inspire students to pursue STEM through driving a basketball scoring robot and helped host monthly free community STEM nights at my High School.